

 This Scribe is only visible to you. Make any changes, then share it.

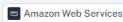
Share



Create AWS VPC, Subnets, Security Group, and EC2 Instance

[+ Add Description](#)

Nhlanhla Malaza 110 steps 20 minutes

 Amazon Web Services  Vocareum

1 Navigate to <https://labs.vocareum.com/main/main.php?m=editor&as...>

In this lab environment, access to AWS services and service actions might be restricted to the ones that you need to complete the lab instructions. You might encounter errors if you attempt to access other services or perform actions beyond the ones that this lab describes.

Accessing the AWS Management Console

- At the top of these instructions, choose [Start Lab](#) to launch this lab.
 - A **Start Lab** panel opens and displays the lab status.
 - Tip** If you need more time to complete the lab, restart the timer for the environment by choosing the [Start Lab](#) button again.
- Wait until you see the message **Lab status: ready**, and then choose the **X** to close the **Start Lab** panel.
- At the top of these instructions, choose [AWS](#).
 - This option opens the AWS Management Console in a new browser tab. The system automatically signs you in.
 - Tip** If a new browser tab does not open, a banner or icon at the top of your browser will indicate that your browser is preventing the site from opening pop-up windows. Choose the banner or icon, and choose **Allow pop-ups**.

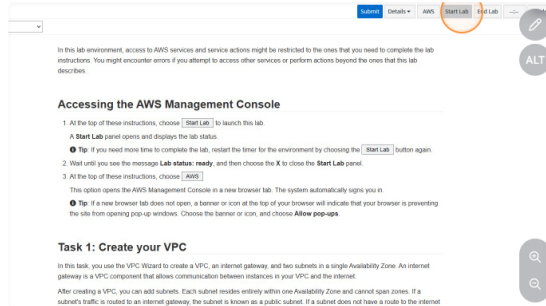
Task 1: Create your VPC

In this task, you use the VPC Wizard to create a VPC, an internet gateway, and two subnets in a single Availability Zone. An internet gateway is a VPC component that allows communication between instances in your VPC and the internet.

After creating a VPC, you can add subnets. Each subnet resides entirely within one Availability Zone and cannot span zones. If a subnet's traffic is routed to an internet gateway, the subnet is known as a public subnet. If a subnet does not have a route to the internet gateway, the subnet is known as a private subnet.

The wizard also creates a NAT gateway, which is used to provide internet connectivity to EC2 instances in private subnets.

2 Click "Start Lab"



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Accessing the AWS Management Console

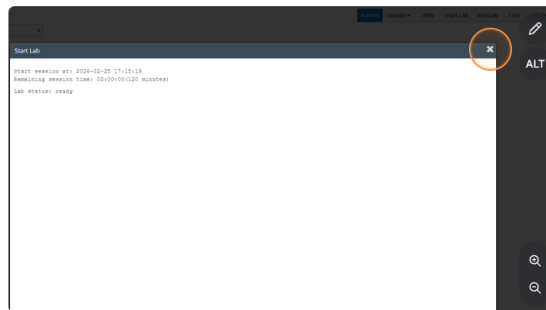
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3 Click "X"

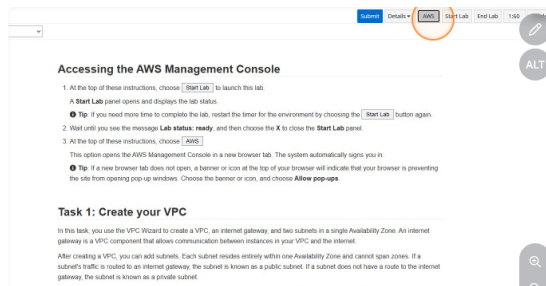


Start Lab

Start: [amazon](#) at: 2024-10-22 17:15:12
Remainder: [amazon](#) time: 02:00:00:00 (3000000)

Lab status: ready

4 Click "AWS"



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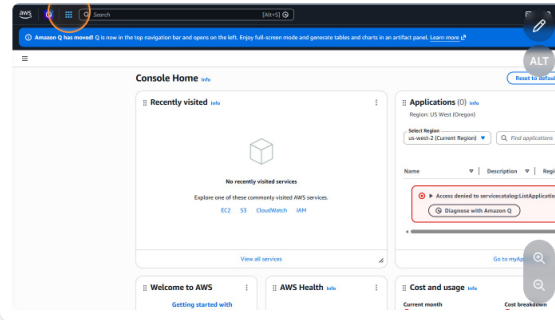
1. Use the search bar in the top navigation bar and type in "AWS". Select the "AWS" link in the search results.
4. At the top right of the instructions, choose **AWS**. The AWS Management Console opens in a new tab.



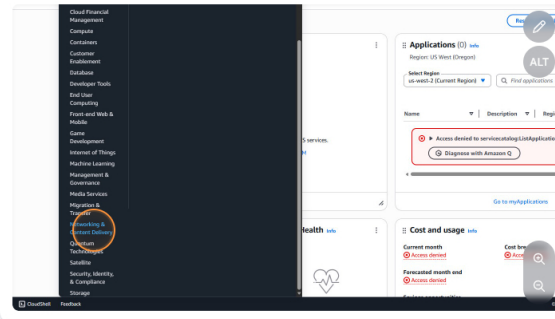
5 Switch to tab Cloud Computing Services - Amazon Web Services (AWS)"



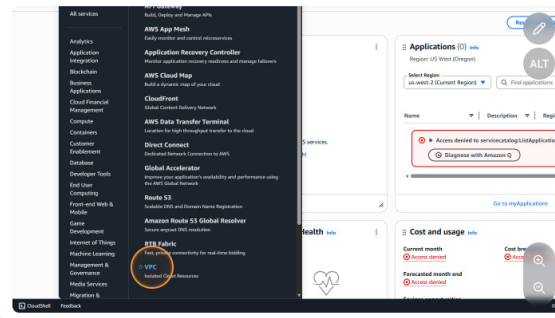
6 Click here.



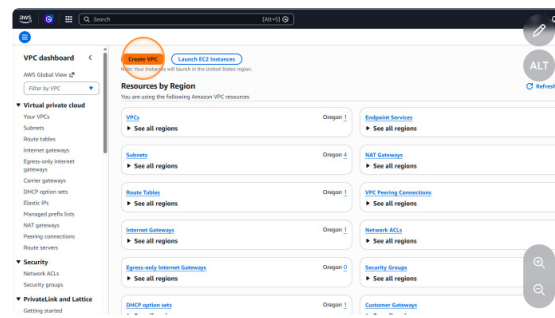
7 Click "Networking & Content Delivery"



8 Click "VPC"



9 Click "Create VPC"



10 Click this checkbox.



A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings

Resources to create [info](#)

Choose only the VPC, Subnet or the VPC and other sub-network resources.

☐ VPC only ☒ VPC and more

Write tag auto-generation [info](#)

Choose the tag for the Name tag. This value will be used to auto-generate Name tags for the VPC.

☒ Auto-generate ☐ Project

IPv4 CIDR block [info](#)

Specify the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 [info](#)

CIDR block size must be between /16 and /28.

IPv4 CIDR block [info](#)

☒ No IPv4 CIDR block ☐ Amazon-provided IPv4 CIDR block

Tenancy [info](#)

Default

► **Encryption settings - optional**



11 Click "1"

Advanced this starting point and the size of your VPC, using a custom CIDR block.

10.0.0.0/16 [info](#)

CIDR block size must be between /16 and /28.

IPv4 CIDR block [info](#)

☒ No IPv4 CIDR block ☐ Amazon-provided IPv4 CIDR block

Tenancy [info](#)

Default

► **Encryption settings - optional**

Number of Availability Zones (AZs) [info](#)

Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 2 3

► **Customize AZs**

Number of public subnets [info](#)

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 1 2

Number of private subnets [info](#)

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 1 2

► **Customize subnets CIDR blocks**

NAT gateways (0) - optional [info](#)

NAT gateways allow private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.

☒ None ☐ Regional ☐ new ☐ Zonal

☒ Introducing regional NAT gateway

AWS now offers a multi-AZ NAT Gateway, eliminating the need for separate NAT Gateways across availability zones.

VPC endpoints [info](#)

Customers can build secure NAT gateway, storage and improve security by connecting to services from the VPC. By default, NAT gateway is used. For more information, see [VPC endpoints](#).



12 Click here.

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13 Click this field.

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Number of public subnets [info](#)

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Number of private subnets [info](#)

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0 1 2

► **Customize subnets CIDR blocks**

Public subnet CIDR block in us-west-2a

10.0.0.0/20 [info](#)

Private subnet CIDR block in us-west-2a

10.0.128.0/20 [info](#)

NAT gateways (0) - optional [info](#)

NAT gateways allow private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.

☒ None ☐ Regional ☐ new ☐ Zonal

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VPC endpoints [info](#)

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14 Type "Backspace 4"



15 Click this field.

► **Customize AZs**

Number of public subnets [info](#)

The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 1 2

Number of private subnets [info](#)

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 1 2

► **Customize subnets CIDR blocks**

Public subnet CIDR block in us-west-2a

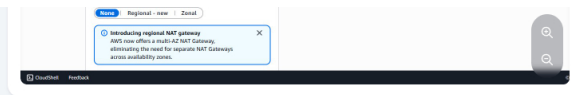
10.0.0.0/20 [info](#)

Private subnet CIDR block in us-west-2a

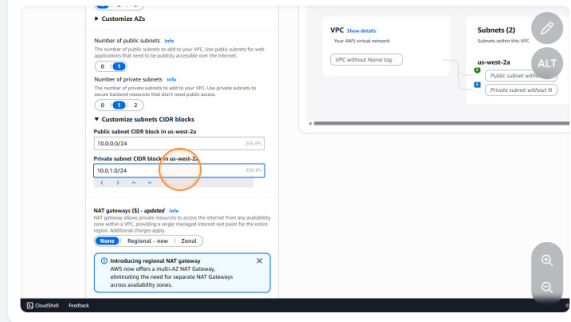
10.0.128.0/20 [info](#)

NAT gateways (0) - optional [info](#)

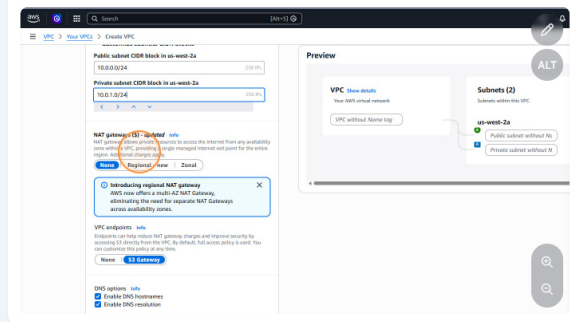
NAT gateways allow private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.



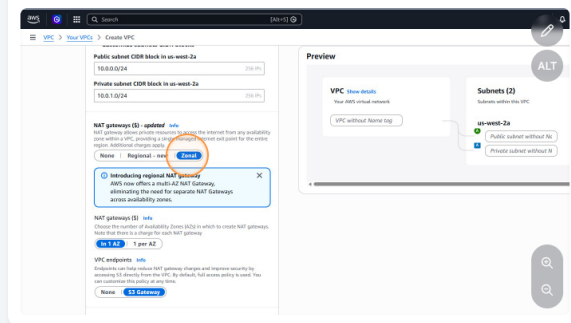
16 Click this field.



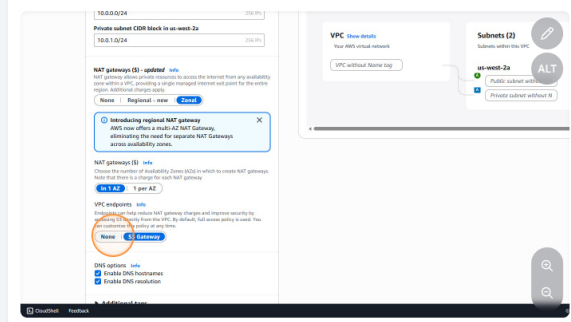
17 Click "NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit po..."



18 Click "Zonal"

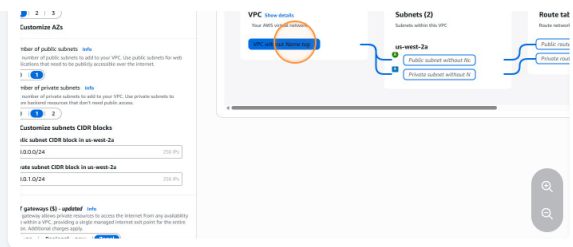


19 Click "None"



20 Click the "VPC without Name tag" field.

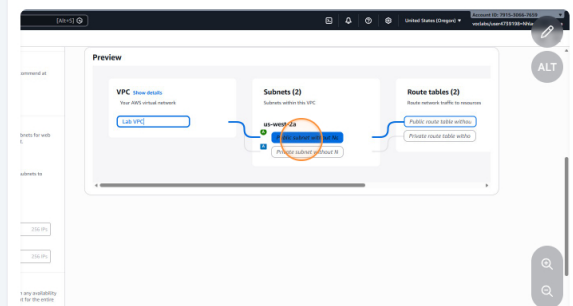




21 Type "Lab VPC"



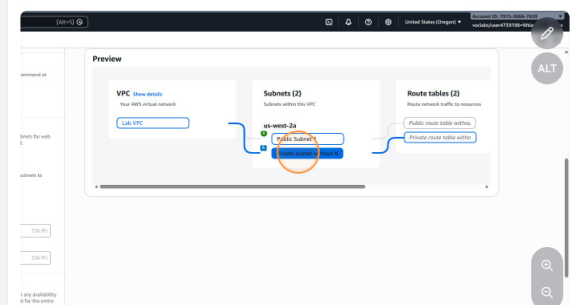
22 Click the "Public subnet without Name tag" field.



23 Type "Public Subnet 1"



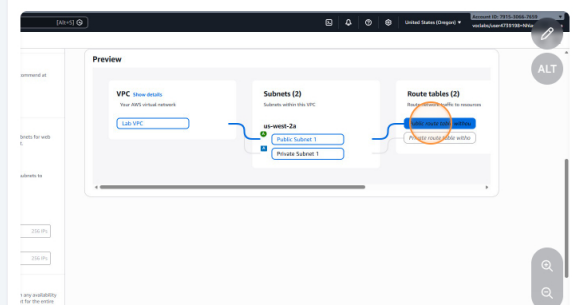
24 Click the "Private subnet without Name tag" field.



25 Type "Private Subnet 1"



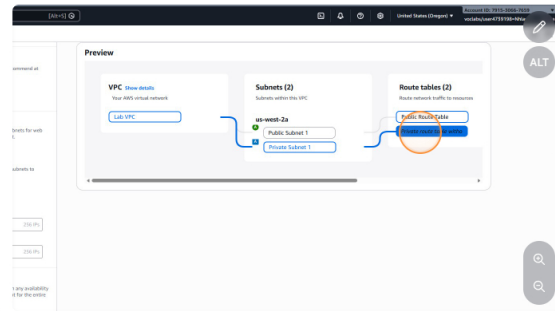
26 Click the "Public route table without Name tag" field.



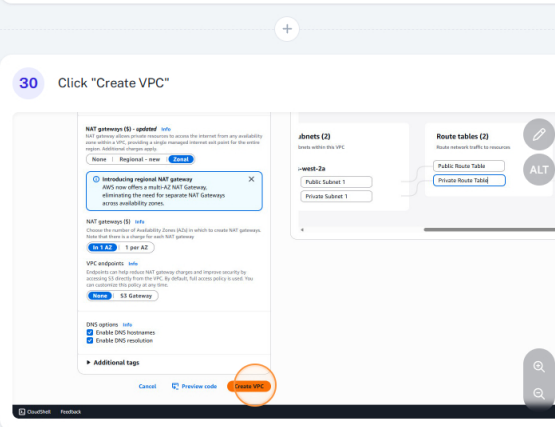
27 Type "Public Route Table"



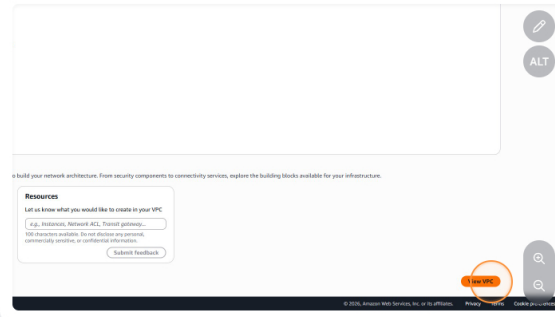
28 Click the "Private route table without Name tag" field.



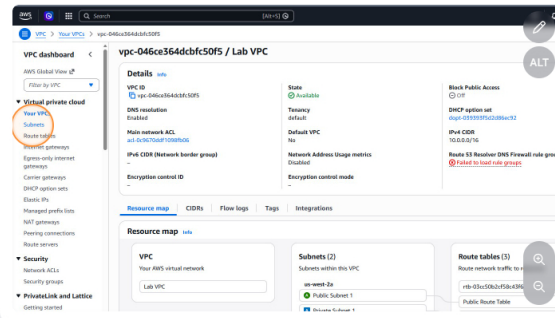
29 Type "Private Route Table"



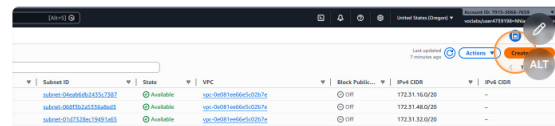
31 Click "View VPC"

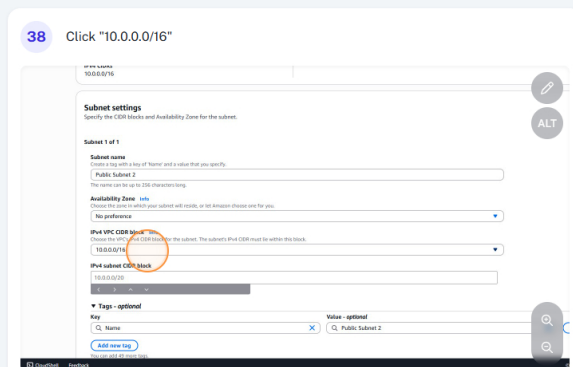
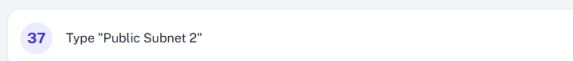
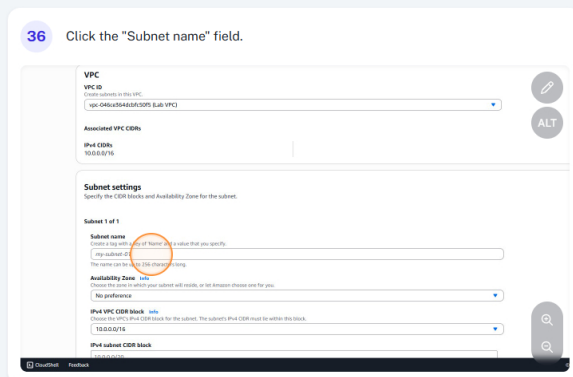
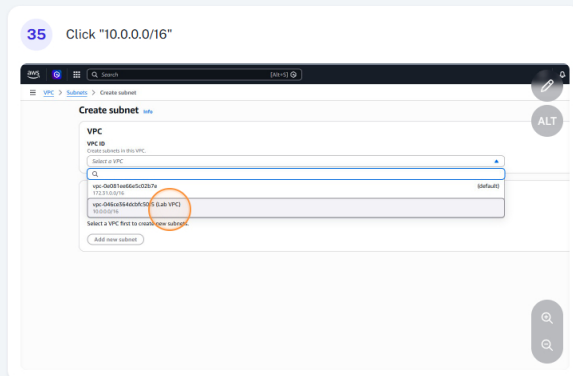
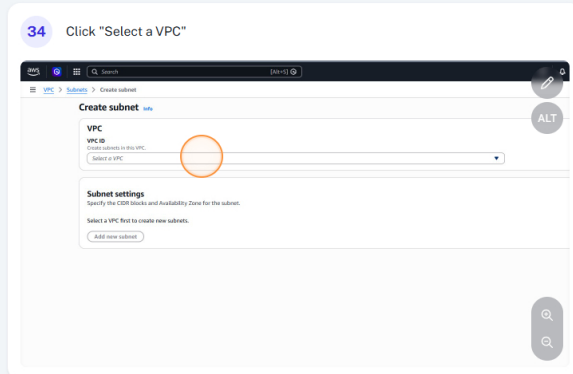
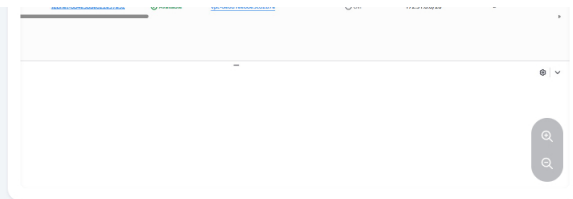


32 Click "Subnets"



33 Click "Create subnet"





39 Click the "10.0.0.0/20" field.

Subnet settings

Subnet 1 of 1

Subnet name

Public Subnet 2

Availability Zone

IPv4 VPC CIDR block

IPv4 subnet CIDR block

Tags - optional

Key

Value - optional

Public Subnet 2

40 Type "10.0.2.0/24"

41 Click "Create subnet"

Subnet settings

Subnet name

Public Subnet 2

Availability Zone

IPv4 VPC CIDR block

IPv4 subnet CIDR block

Tags - optional

Key

Value - optional

Public Subnet 2

Create subnet

42 Click "Create subnet"

Subnet settings

Subnet name

Public Subnet 2

Availability Zone

IPv4 VPC CIDR block

IPv4 subnet CIDR block

Tags - optional

Key

Value - optional

Public Subnet 2

Create subnet

43 Click "Select a VPC"

Subnet settings

Subnet name

Public Subnet 2

Availability Zone

IPv4 VPC CIDR block

IPv4 subnet CIDR block

Tags - optional

Key

Value - optional

Public Subnet 2

Create subnet

44 Click "vpc-046ce364dcbfc50f5 (Lab VPC)"

Subnet settings

Subnet name

Public Subnet 2

Availability Zone

IPv4 VPC CIDR block

IPv4 subnet CIDR block

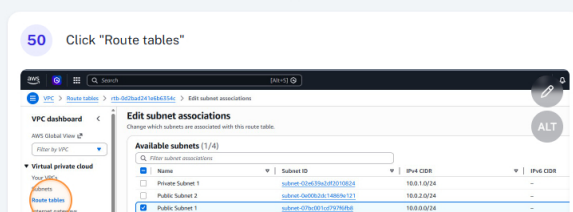
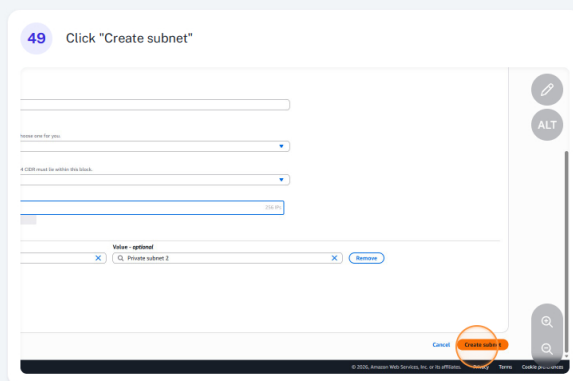
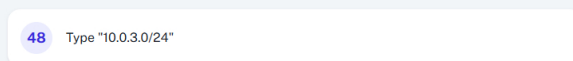
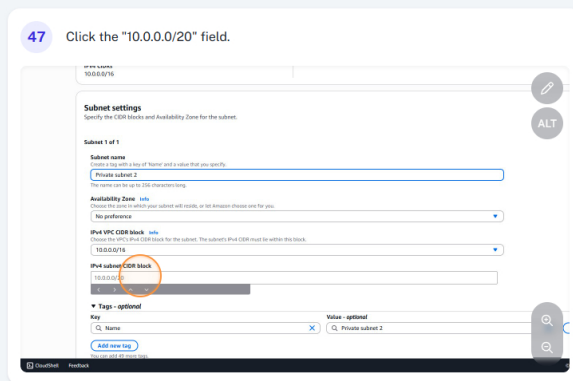
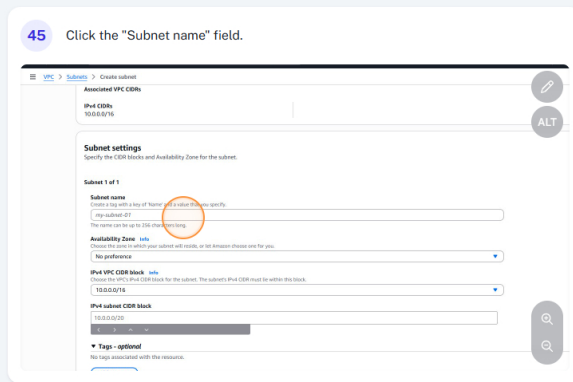
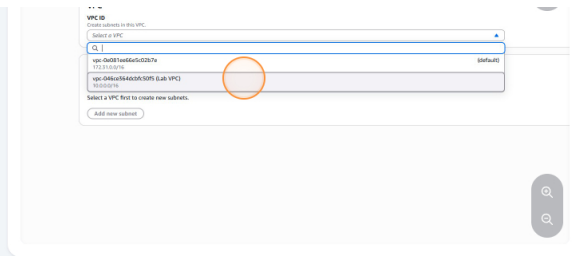
Tags - optional

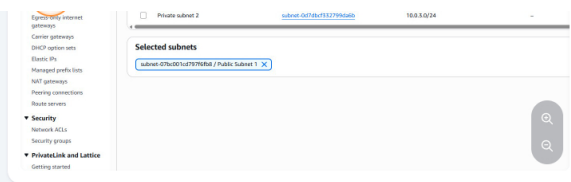
Key

Value - optional

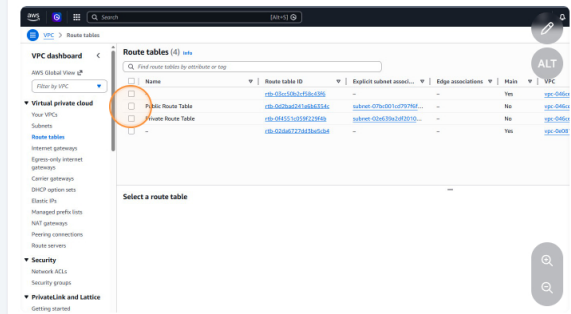
Public Subnet 2

Create subnet

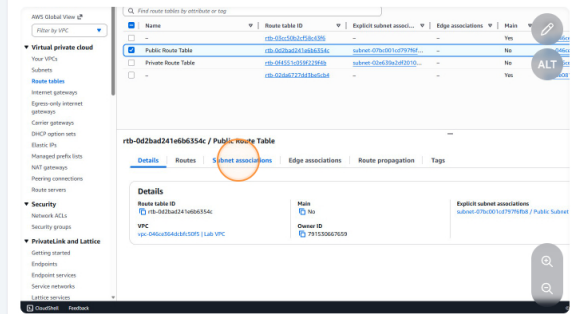




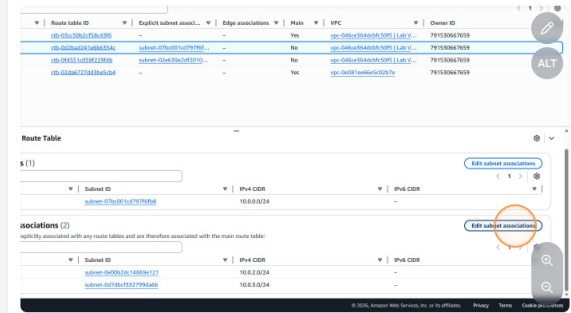
51 Click this checkbox.



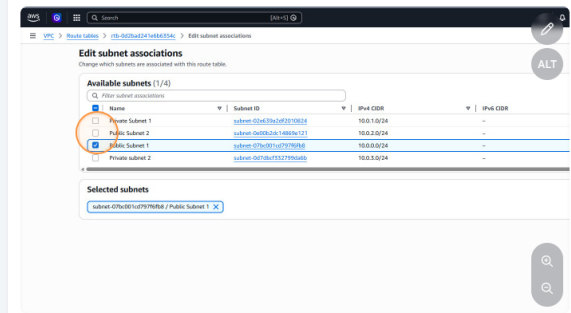
52 Click "Subnet associations"



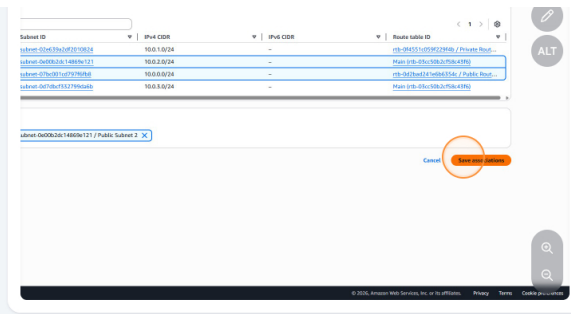
53 Click "Edit subnet associations"



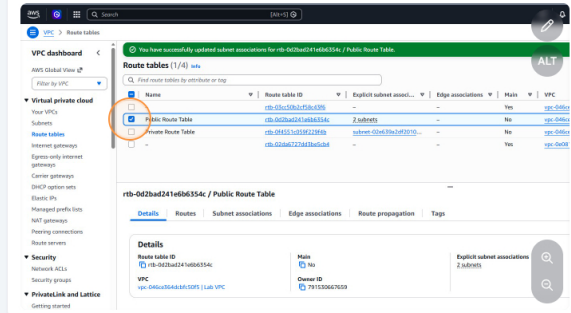
54 Click this checkbox.



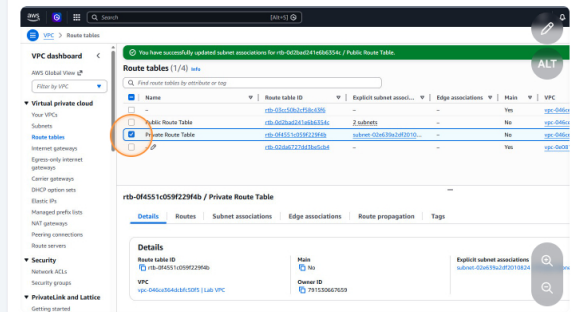
55 Click "Save associations"



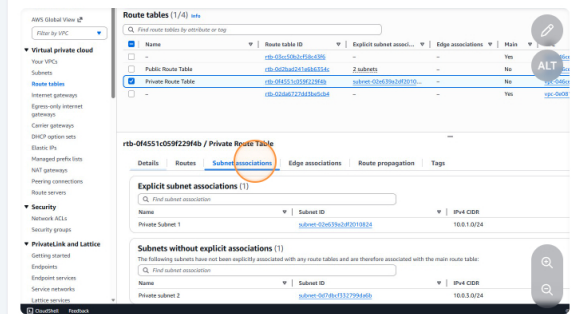
56 Click this checkbox.



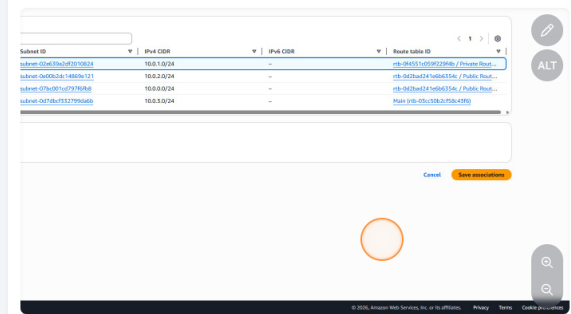
57 Click this checkbox.



58 Click "Subnet associations"



59 Click "Edit subnet associations"



60 Click this checkbox.

Available subnets (1/4)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
Private Subnet 1	subnet-03a659a2f2010824	10.0.1.0/24	-
Public Subnet 2	subnet-0a0b62a14869a121	10.0.2.0/24	-
Public Subnet 1	subnet-07ba0071d79796a8	10.0.0.0/24	-
Private subnet 2	subnet-0a78db1532799a4b	10.0.3.0/24	-

Selected subnets

subnet-03a659a2f2010824 / Private Subnet 1

61 Click "Save associations"

Save associations

62 Click "Security groups"

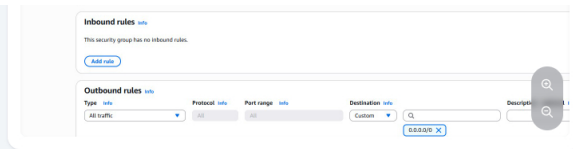
Security groups

63 Click "Create security group"

Create security group

64 Click the "Security group name Info" field.

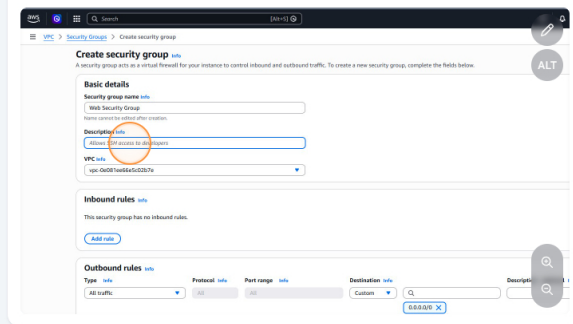
Security group name



65 Type "Web Security Group"



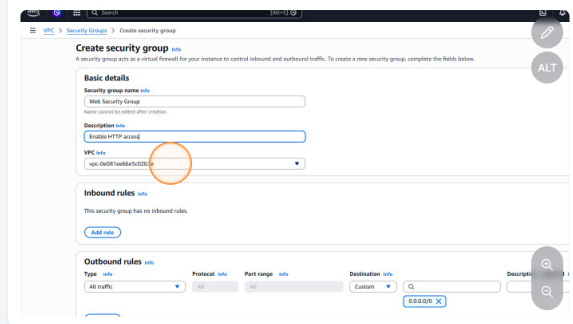
66 Click the "Description Info:" field.



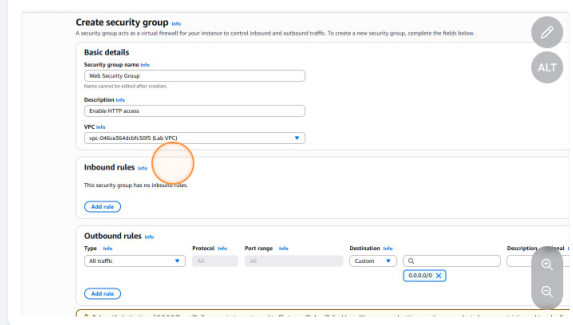
67 Type "Enable HTTP access"



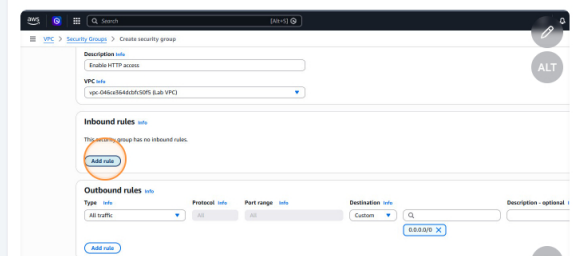
68 Click "vpc-0e081ee66e5c02b7e"



69 Click "vpc-046ce364dcbfc50f5 (Lab VPC)"



70 Click "Add rule"



Rules with destination of 0.0.0.0/0 or /0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific IP addresses.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

71 Click "Custom TCP"

Security Group

Create security group

Enable HTTP access

vpc-045c5d46d05f505f Sub VPC

Inbound rules

Type

Custom TCP

Protocol

TCP

Port range

0

Source

Custom

Description - optional

Outbound rules

Type

All traffic

Protocol

All

Port range

All

Destination

Custom

Description - optional

Rules with destination of 0.0.0.0/0 or /0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific IP addresses.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

72 Click "HTTP"

Security Group

Create security group

Enable HTTP access

vpc-045c5d46d05f505f Sub VPC

Inbound rules

Type

Custom TCP

Protocol

TCP

Port range

0

Source

Custom

Description - optional

Outbound rules

Type

All traffic

Protocol

All

Port range

All

Destination

Custom

Description - optional

Rules with destination of 0.0.0.0/0 or /0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific IP addresses.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

73 Click "Custom"

Security Group

Create security group

Enable HTTP access

vpc-045c5d46d05f505f Sub VPC

Inbound rules

Type

Custom TCP

Protocol

TCP

Port range

0

Source

Custom

Description - optional

Outbound rules

Type

All traffic

Protocol

All

Port range

All

Destination

Custom

Description - optional

Rules with destination of 0.0.0.0/0 or /0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific IP addresses.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

74 Click "Anywhere-IPv4"

Security Group

Create security group

Enable HTTP access

vpc-045c5d46d05f505f Sub VPC

Inbound rules

Type

Custom TCP

Protocol

TCP

Port range

0

Source

Anywhere

Description - optional

Outbound rules

Type

All traffic

Protocol

All

Port range

All

Destination

Custom

Description - optional

Rules with destination of 0.0.0.0/0 or /0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific IP addresses.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

75 Click this text field.

Security Group

Create security group

Enable HTTP access

vpc-045c5d46d05f505f Sub VPC

Inbound rules

Type

Custom TCP

Protocol

TCP

Port range

0

Source

Anywhere

Description - optional

Outbound rules

Type

All traffic

Protocol

All

Port range

All

Destination

Custom

Description - optional

Rules with destination of 0.0.0.0/0 or /0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific IP addresses.

Tags - optional
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

The screenshot shows the AWS IAM console's 'Security rules' tab for an instance profile. The 'Part range' is set to 'All' and the 'Destination' is 'Custom' with the value '0.0.0.0/0'. The 'Description' is 'optional info'. The 'Security rules' list shows two rules: 'Allow access to your instance' and 'Deny access to traffic to any IPv4 or IPv6 address'.

76 Type "Permit web requests"

Part range

Source

Anywhere

Description: optional

Force rules required

Delete

as to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Part range

Destination

Custom

Description: optional

Delete

crises to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

< controls of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Create rule

Go to next step >

78 Click this icon.

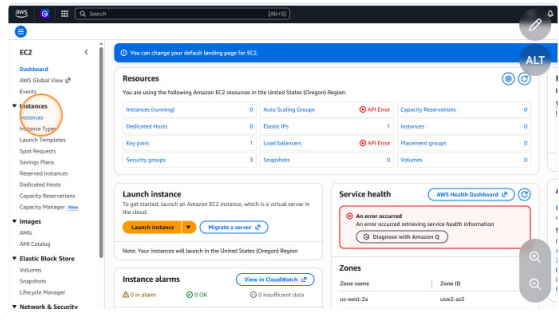
The screenshot shows the AWS VPC console interface. At the top, the breadcrumb navigation indicates the path: VPC > Security groups > ig-0a9384adecdf2d6d4c. The main header area shows the group's name 'ig-0a9384adecdf2d6d4c' and a status message: 'Security group ig-0a9384adecdf2d6d4c: 1 VPC Security Group was created successfully'. Below this, the 'Details' tab is active, displaying the group's ID and a list of rules. The 'Rules' tab is selected, showing a single inbound rule for port 22 (SSH) with a source of 'Anywhere'. The 'Rules' tab is also highlighted in the left-hand navigation pane.

79 Click "Compute"

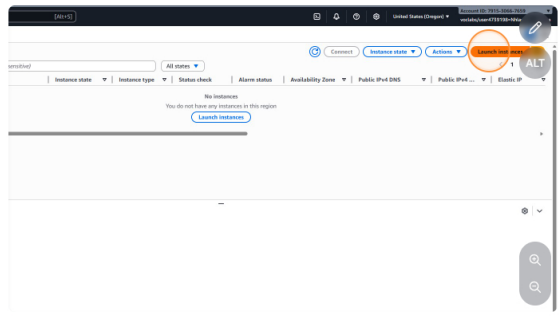
The screenshot shows the AWS IAM console interface. On the right, the 'Groups' table lists 'IAMUsersGroup' with a 'Users' link. The left-hand navigation pane is expanded, showing categories like 'VPC dashboard', 'Security', and 'PrivateLink'. The 'Security' category is highlighted, and 'IAMUsersGroup' is selected under the 'Groups' section.

81 Switch to tab `us-west-2.console.aws.amazon.com/ec2/home?region=u...`

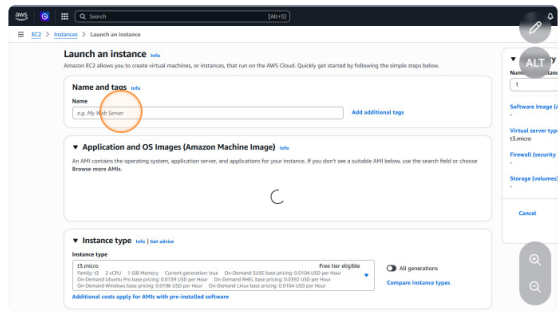
82 Click "Instances"



83 Click "Launch instances"

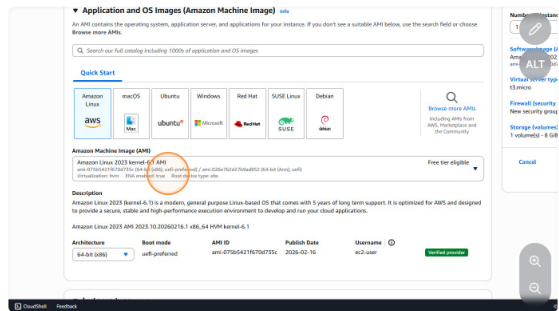


84 Click the "Name" field.



85 Type "Web Server 1"

86 Click "ENA enabled: true"



- 87 Click "Launch an instance InfoAmazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by followi..."

Web Server 1 Add additional tags

Search

Amazon Linux 2 LTS with SQL Server 2017 Standard

Free tier eligible

Description

Amazon Linux 2023 (Barnard 4.0) is a medium, general purpose Linux-based OS that comes with 5 years of long-term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.30202310.1 x86_64 HVM kernel 6.1

Architecture: x86_64 (SAGE) Boot mode: uefi-grubefontd AMI ID: ami-0750542176707731c Publish Date: 2023-02-16 Username: ec2-user

Verified resource

Cancel

- 88 Click "Select"

to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.30202310.1 x86_64 HVM kernel 6.1

Architecture: x86_64 (SAGE) Boot mode: uefi-grubefontd AMI ID: ami-0750542176707731c Publish Date: 2023-02-16 Username: ec2-user

Verified resource

Instance type info Set advice

Instance type

Amazon Linux 2 LTS with SQL Server 2017 Standard

Free tier eligible

Additional costs apply for AMIs with pre-installed software

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name: required

Default

Create new key pair

Network settings info

Network: vpc-0d081ee6e5c02b7e

Edit

Cancel

- 89 Click "vockey"

to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.30202310.1 x86_64 HVM kernel 6.1

Architecture: x86_64 (SAGE) Boot mode: uefi-grubefontd AMI ID: ami-0750542176707731c Publish Date: 2023-02-16 Username: ec2-user

Verified resource

Instance type info Set advice

Instance type

Amazon Linux 2 LTS with SQL Server 2017 Standard

Free tier eligible

Additional costs apply for AMIs with pre-installed software

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name: required

Default

Create new key pair

Network settings info

Network: vpc-0d081ee6e5c02b7e

Edit

Cancel

- 90 Click "Edit"

to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.30202310.1 x86_64 HVM kernel 6.1

Architecture: x86_64 (SAGE) Boot mode: uefi-grubefontd AMI ID: ami-0750542176707731c Publish Date: 2023-02-16 Username: ec2-user

Verified resource

Instance type info Set advice

Instance type

Amazon Linux 2 LTS with SQL Server 2017 Standard

Free tier eligible

Additional costs apply for AMIs with pre-installed software

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name: required

Default

Create new key pair

Network settings info

Network: vpc-0d081ee6e5c02b7e

Edit

Cancel

- 91 Click "(default)"

to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.30202310.1 x86_64 HVM kernel 6.1

Architecture: x86_64 (SAGE) Boot mode: uefi-grubefontd AMI ID: ami-0750542176707731c Publish Date: 2023-02-16 Username: ec2-user

Verified resource

Instance type info Set advice

Instance type

Amazon Linux 2 LTS with SQL Server 2017 Standard

Free tier eligible

Additional costs apply for AMIs with pre-installed software

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name: required

Default

Create new key pair

Network settings info

Network: vpc-0d081ee6e5c02b7e

Edit

Cancel

Enable

Firewall (security group)

A security group is a set of firewall rules that control the traffic to your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

search-wizard-1

92 Click "10.0.0.0/16"

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

Create new key pair

Network settings

VPC - required

vpc-0a0c5646b0a095 Sub VPC

Create new VPC

Subnet

10.0.0.0/16

vpc-0a0c5646b0a095 Sub VPC

Create new subnet

10.0.0.0/16

vpc-0a0c5646b0a095 Sub VPC

Create new subnet

Auto-assign public IP

Enable

Enable additional addresses

Firewall (security group)

A security group is a set of firewall rules that control the traffic to your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

search-wizard-1

93 Click "VPC: vpc-046ce364dbcf50f5"

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

Create new key pair

Network settings

VPC - required

vpc-046ce364dbcf50f5 Sub VPC

Create new VPC

Subnet

subnet-02d35b20f010824

vpc-046ce364dbcf50f5 Sub VPC

Create new subnet

subnet-02d35b20f010824

vpc-046ce364dbcf50f5 Sub VPC

Create new subnet

Auto-assign public IP

Disable

Enable additional addresses

Firewall (security group)

A security group is a set of firewall rules that control the traffic to your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

search-wizard-1

94 Click "Public Subnet 2"

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

Create new key pair

Network settings

VPC - required

vpc-046ce364dbcf50f5 Sub VPC

Create new VPC

Subnet

subnet-02d35b20f010824

vpc-046ce364dbcf50f5 Sub VPC

Create new subnet

subnet-02d35b20f010824

vpc-046ce364dbcf50f5 Sub VPC

Create new subnet

Auto-assign public IP

Disable

Enable additional addresses

Firewall (security group)

A security group is a set of firewall rules that control the traffic to your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

search-wizard-1

95 Click "Disable"

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

Create new key pair

Network settings

VPC - required

vpc-046ce364dbcf50f5 Sub VPC

Create new VPC

Subnet

subnet-02d35b20f010824

vpc-046ce364dbcf50f5 Sub VPC

Create new subnet

subnet-02d35b20f010824

vpc-046ce364dbcf50f5 Sub VPC

Create new subnet

Auto-assign public IP

Disable

Enable additional addresses

Firewall (security group)

A security group is a set of firewall rules that control the traffic to your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

search-wizard-1

96 Click "Enable"

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

vockey

Create new key pair

Network settings

VPC - required

vpc-046ce364dbcf50f5 Sub VPC

Create new VPC

Subnet

subnet-02d35b20f010824

vpc-046ce364dbcf50f5 Sub VPC

Create new subnet

subnet-02d35b20f010824

vpc-046ce364dbcf50f5 Sub VPC

Create new subnet

Auto-assign public IP

Enable

Enable additional addresses

Firewall (security group)

A security group is a set of firewall rules that control the traffic to your instance. Add rules to allow specific traffic to reach your instance.

Create security group

Select existing security group

Security group name - required

search-wizard-1

☐ Create security group

☐ Select existing security group

Security group name - required
 search-wizard-1

Description - required | info
 search-wizard-1 created 2020-02-26T01:54:44.842Z

Inbound Security Group Rules
 Security group rule 1 (TCP: 22, 0.0.0.0/0)

Type | info | Protocol | info | Port range | info

TCP

97 Click "Select existing security group"

☐ Create security group

☐ Select existing security group

Security group name - required
 search-wizard-1

Description - required | info
 search-wizard-1 created 2020-02-26T01:54:44.842Z

Inbound Security Group Rules
 Security group rule 1 (TCP: 22, 0.0.0.0/0)

Type | info | Protocol | info | Port range | info

TCP

98 Click "Select security groups"

☐ Create security group

☐ Select existing security group

Common security groups | info
 Select security group

Compare security group rules

Configure storage | info
 1x 8 GB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

99 Click "VPC: vpc-046ce364dcbfc50f5"

☐ Create security group

☐ Select existing security group

Common security groups | info
 Select security group

Compare security group rules

Configure storage | info
 1x 8 GB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

100 Click "Advanced network configuration"

☐ Create security group

☐ Select existing security group

Common security groups | info
 Select security group

Compare security group rules

Advanced network configuration

Configure storage | info
 1x 8 GB gp3 Root volume, 3000 IOPS, Not encrypted

Add new volume

Click refresh to view backup information
 The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 File systems

101 Switch to tab Workbench -Vocareum"

102 Click "service httpd start"

• **Amazon Machine Image (AMI):** From dropdown, choose Amazon Linux 2 AMI (HVM)

- In the **Instance type** section, choose **t3.micro**
- In the **Key pair (login)** section, choose **nokey**

36. In the **Network settings** section, choose **Elastic** and configure the following options:

- **VPC:** **required**. Choose **Lab VPC**
- **Subnet:** Choose **Public Subnet 2**
- **Auto-assign public IP:** Choose **Enable**
- **Firewall (security groups):** Choose **Select existing security group**
 - Choose **Web Security Group**

37. Expand **Advanced details**

38. Under **User data**, copy and paste the following code

```
#!/bin/bash
#Install Apache web server and PHP
yum install -y httpd mysql php
#Download Lab files
wget https://aws-ric-lambdaobjects.s3.us-west-2.amazonaws.com/C68-1P-380-863387-1/267-Lab-WF-hu1d-rgc-web-server/v3/Tab-asp.php
unzip Lab-web-asp-v3 /var/www/html/
#Start on web server
chkconfig httpd on
service httpd start
```

39. Choose **Launch instance**

40. To display the launched instance, choose **View all instances**.

41. Wait until the **Web Server 1** shows **22 checks passed** in the **Status check** column.

103 Click here.

Select Select Select

Elastic Express [info](#) **Elastic Express UDP** [info](#) **Elastic Express** [info](#)

The selected instance type does not support multiple network cards. The selected instance type does not support Elastic Express. The selected instance type does not support Elastic Express.

Info connection tracking timeout [info](#)

☐ Enable

[Add network interface](#)

Configure storage [info](#) [Advanced](#)

1x 8 GB gp3 Root volume, 5000 IOPS, Not encrypted

[Add new volume](#)

☐ Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0x File systems [Edit](#)

[Advanced details](#) [info](#)

[Cancel](#)

104 Click the "User data-optional" field.

Metadata response hop limit [info](#)

2

Allow tags in metadata [info](#)

Select

User data-optional [info](#)

Optional. Use this user data or enter it in the field.

[Choose file](#)

☐ User data has already been base64 encoded

[Cancel](#)